CornellBowers
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"Where Should I Look?"

Reimplementing Visual Attention for Neural Image Captioning

Based on "Show, Attend, and Tell" by (Xu et al., ICML 2015)

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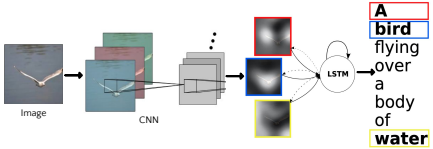
Paper - Xu et al.

How do we get a computer to look at an image and describe it in natural language?

Earlier Models: **Problem:** Bottleneck -> Generic Captions

Paper's Idea: ATTENTION



Key Ideas:

Grid of Features: Model represents the image as **multiple spatial regions** for spatial reasoning

Attention: For each word the model assigns weights to **all regions** to compute weighted combination

Soft Attention: uses all regions with a weighted average

Hard Attention: selects on region but was found to be harder to train (through RL via REINFORCE)

A stop sign is on a road with a mountain in the background.

Our Approach:

We trained 3 different attention models to see if attention actually understanding images or just helping the model compute better features?

- Baseline: Standard soft attention from the paper
- Shuffled Attention: Same attention weights but in random locations
- Sharpened Attention: Encourage attention to be more concentrated

We isolate attention as the only changing component so we can directly test its role in the model.

The Attention Mechanism

Intuition: At each word, the model asks: which image regions matter most right now?

Attention Module

$$e_{ti} = f_{\text{att}}(\mathbf{a}_i, \mathbf{h}_{t-1})$$

$$\alpha_{ti} = \frac{\exp(e_{ti})}{\sum_{k=1}^{L} \exp(e_{tk})}.$$

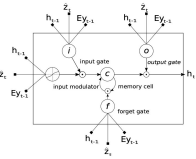
$$\mathbb{E}_{p(s_t|a)}[\hat{\mathbf{z}}_t] = \sum_{i=1}^{L} \alpha_{t,i}\mathbf{a}_i$$

Alignment Score: Score each spatial region (each a_i is a CNN feature vector describing a 16×16 patch) against the previous hidden state h_{t-1} to determine relevance

Attention Weights: Normalize to get a distribution over all locations

Soft Attention Context Vector: Weighted combination → what the decoder 'sees' at this step

Cross Attention: query comes from the text decoder and the keys/values come from the image encoder



Forget Gate (f): Sees where attention is looking now; clears "dog" when attention shifts to "background"

Input Gate (i): Uses $\hat{z}_t$ to write the currently attended visual content into memory.

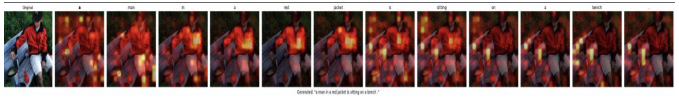
Memory cell (c): Accumulates visual information from attention over multiple timesteps

Output gate (o): Filters memory into h_t which serves two roles: predicts the next word and tells attention where to look next

Attention In Action

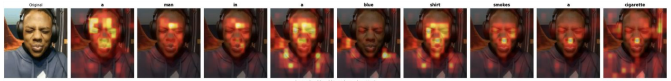
Caption: "A man in a red jacket is sitting on a bench"

Model attends to the person when generating "man" or "jacket" and the broader scene for spatial words like "bench"



Caption: "A man in a blue shirt smokes a cigarette"

The model attends to the face region throughout but misidentifies the action (smokes cigarette). The decoder overfits to common co-occurrences in training data.

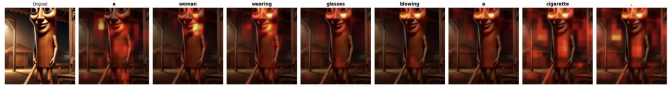


Caption: "A man in a yellow shirt is standing in the middle of a crowd"

The model attends to the correct object for content words (man, yellow, shirt) but spreads attention diffusely for function words (a, the, in, is). This is seen commonly in both our results and that of the paper.

Caption: "A woman wearing glasses blowing a cigarette"

Hard for model to reason about abstract images different from training data



Datasets

Dataset	Images	Caps/Img	Role
Flickr 8k	8,000	5	Dev / Debug
Flickr30k	31,000	5	Primary eval

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References:
  - a dog jumping from another dog .
  - one dog is jumping in the air and the other is standing next to it .
  - the two dogs are jumping around .
  - two dogs are playing .
  - two large brown and white dogs are playing in a grassy area .
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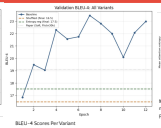
BLEU: Measures n-gram overlap between generated and reference captions. BLEU-1 checks single-word matches. BLEU-4 checks 4-word phrase matches → higher is harder

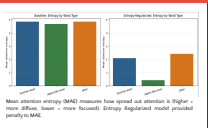
METEOR: Matches words using synonyms and stems, not just exact matches. Correlates better with human judgment than BLEU

Results

Training and Validation per Variant



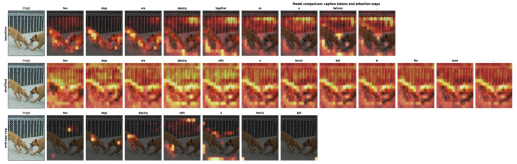
BLEU-4 Scores Per Variant



Mean attention entropy (MAE) measures how spread out attention is (higher = more diffuse, lower = more focused). Entropy Regularized model provided penalty to MAE.

Model	BLEU-1	BLEU-2	BLEU-3	BLEU-4	METEOR
Paper (Soft)	66.7	43.4	28.8	19.1	18.49
Our (Baseline)	**62.1**	**41.9**	**29.0**	**20.1**	**20.85**
Shuffled Attention	58.1	37.3	24.6	16.5	19.05
Entropy Regulated	58.3	38.4	25.8	17.5	19.09

Comparison against the paper's reported Flickr 30k scores. Our soft attention model achieves a BLEU-4 of 20.1 and METEOR of 20.85, exceeding the paper's METEOR (18.49) while falling slightly short on BLEU-1 (62.1 vs 66.7). The other models fall behind the paper for all except METEOR.



Baseline (top): Two dogs are playing together on a balcony; Shuffled: Two dogs are playing with a tennis ball in the snow; Entropy Regularized: Two dogs playing with a tennis ball

Takeaways & Future Work

Takeaways:

- The model attends correctly to objects for content words but spreads diffusely for function words
- Failure cases stem from the decoder relying on common training co-occurrences rather than from attention looking at the wrong region
- Our model finds it difficult to reason about images that are very different from training data

Future Work:

- Hard attention via REINFORCE to learn the context vector via policy gradient
- Modern encoders (ResNet-101, ViT)
- Scale to MS COCO (82K images)

References: Xu, Kelvin, et al. "Show, attend and tell: Neural image caption generation with visual attention." International conference on machine learning. PMLR, 2015.